

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton (NeurIPS 2012)

Reviewed by BeomSoo Son

Background & Problem Statement

Limitations of Small Datasets

Previous datasets are insufficient to capture the considerable variability of objects in realistic settings.

Emergence of Large-Scale Data

Introduction of ImageNet: Over 15 million labeled high-resolution images belonging to roughly 22,000 categories.

New Requirement

Learning from millions of images requires a model with an immense "learning capacity".

Limitations & The Breakthrough

The Dilemma of CNN

Despite their attractive qualities, CNNs have been prohibitively expensive to apply in large scale to high-resolution images.

Key Contributions

GPU Parallelization: Wrote a highly-optimized GPU implementation of 2D convolution to facilitate training.

Preventing Overfitting: Employed several effective techniques (data augmentation, "dropout") in a massive network with 60 million parameters.

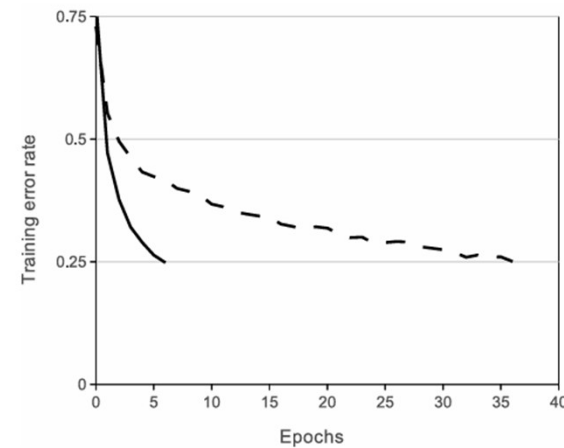
Record-Breaking Performance: Best results ever reported on ILSVRC-2010 and 2012 datasets.

ReLU Nonlinearity

- Uses a non-saturating nonlinearity $f(x) = \max(0, x)$ instead of traditional saturating models like tanh.
- Deep convolutional neural networks with ReLUs train several times faster than their equivalents with tanh units.

Quantitative Evidence

On CIFAR-10 with a 4-layer CNN, the ReLU network reaches 25% training error **6× faster** than the equivalent tanh network.



Training on Multiple GPUs

- Spreads the network across two GPUs to overcome the 3GB memory limit of a single GPU.
- GPUs communicate only in certain layers to optimize the amount of communication and computation.

Connectivity Rule

Only the 3rd conv layer is connected to all kernel maps of the previous layer; the 4th and 5th conv layers are connected only to kernel maps on the same GPU.

Quantitative Effect

Reduces top-1 error by 1.7% and top-5 error by 1.2%, compared with a one-GPU network using half as many kernels.

Local Response Normalization (LRN)

- Implements a form of lateral inhibition inspired by real neurons to aid generalization.
- Creates competition for big activities amongst neuron outputs computed using different kernels.

Quantitative Effect

Applying LRN reduces top-1 error by 1.4% and top-5 error by 1.2%.

Overlapping Pooling

- Pooling layers summarize neighborhoods of size 3×3 spaced 2 pixels apart ($s < z$).
- Models with overlapping pooling find it slightly more difficult to overfit compared to traditional non-overlapping schemes.

Quantitative Effect

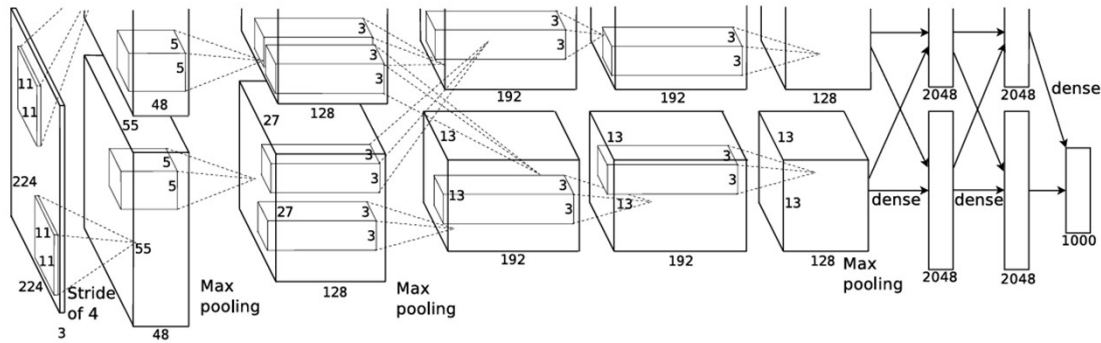
Introducing overlapping pooling reduces top-1 error by 0.4% and top-5 error by 0.3%.

Overall Architecture

8 Learned Layers: 5 convolutional layers + 3 fully-connected layers.

Massive Scale: 60 million parameters, 650,000 neurons. Input image: 224x224x3.

레이어	커널/필터 크기	개수	Stride
Conv1	11x11x3	96	4
Conv2	5x5x48	256	1
Conv3	3x3x256	384	1
Conv4	3x3x192	384	1
Conv5	3x3x192	256	1
FC6/7/8	-	4096 / 4096 / 1000	-



Data Augmentation

[Training Time] Random Cropping & Flipping

Generates image translations and horizontal reflections by extracting random 224×224 patches from 256×256 images.

[Training Time] PCA Color Augmentation

Alters the intensities of RGB channels in training images using Principal Component Analysis (PCA).

[Test Time] 10-Patch Averaging

At test time, five 224×224 patches (four corners + center) plus their horizontal reflections — 10 patches in total — are extracted, and the softmax outputs are averaged for the final prediction.

Dropout

- Sets the output of each hidden neuron to zero with a probability of 0.5.
- Reduces complex co-adaptations of neurons, forcing the network to learn more robust features.
- Employed in the first two fully-connected layers to combat substantial overfitting.

Details of Learning

Optimization Setup

SGD with batch size 128, momentum 0.9, weight decay 0.0005.

Weight decay acts not merely as a regularizer, but reduces the model's training error.

Weight Initialization

Weights are initialized from a zero-mean Gaussian distribution with standard deviation 0.01.

Bias Initialization

2nd, 4th, 5th conv layers and FC layers biases initialized to constant 1 (accelerates early learning with ReLU).

Learning Rate Schedule

Learning rate starts at 0.01 and is divided by 10 whenever validation error plateaus (reduced 3 times in total).

Training Time

~90 epochs on 1.2M images, 5-6 days on two NVIDIA GTX 580 3GB GPUs.

Quantitative Results (ILSVRC)

ILSVRC-2010

모델	Top-1 오류율	Top-5 오류율
이전 SOTA	47.1%	28.2%
AlexNet	37.5%	17.0%

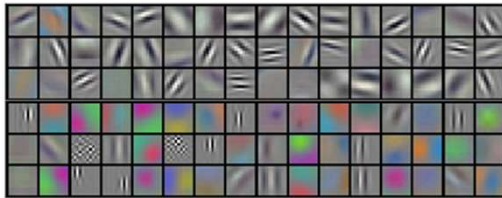
ILSVRC-2012

모델	Top-5 오류율
2등 팀	26.2%
AlexNet 단일 모델	18.2%
AlexNet 7-CNN 앙상블	15.3%

Qualitative Evaluations

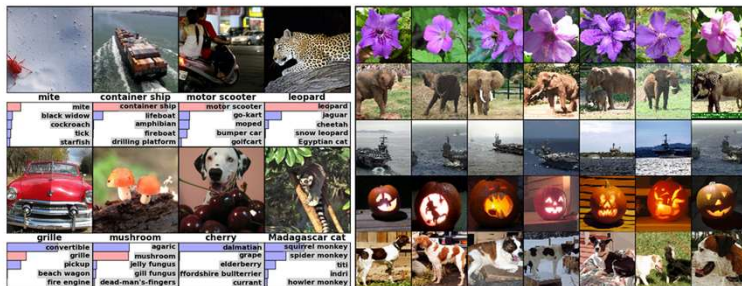
Learned Convolutional Kernels

- The first convolutional layer learned frequency- and orientation-selective kernels, as well as various colored blobs.
- Kernels on GPU 1 are largely color-agnostic, while kernels on GPU 2 are largely color-specific.



Robust Object Recognition

- The network is capable of recognizing even off-center objects in test images.
- When incorrect, top-5 predicted labels still appear visually and semantically reasonable.



Conclusion

Triumph of Deep CNNs

A large, deep CNN can achieve record-breaking results on a highly challenging dataset using purely supervised learning.

The Importance of Depth

Performance degrades if a single convolutional layer is removed (~2% loss in top-1 performance), proving depth is important.

Future Directions

Authors aim to apply very large and deep CNNs on video sequences, where temporal structure provides helpful information.

Thank You